

PH101: 2023

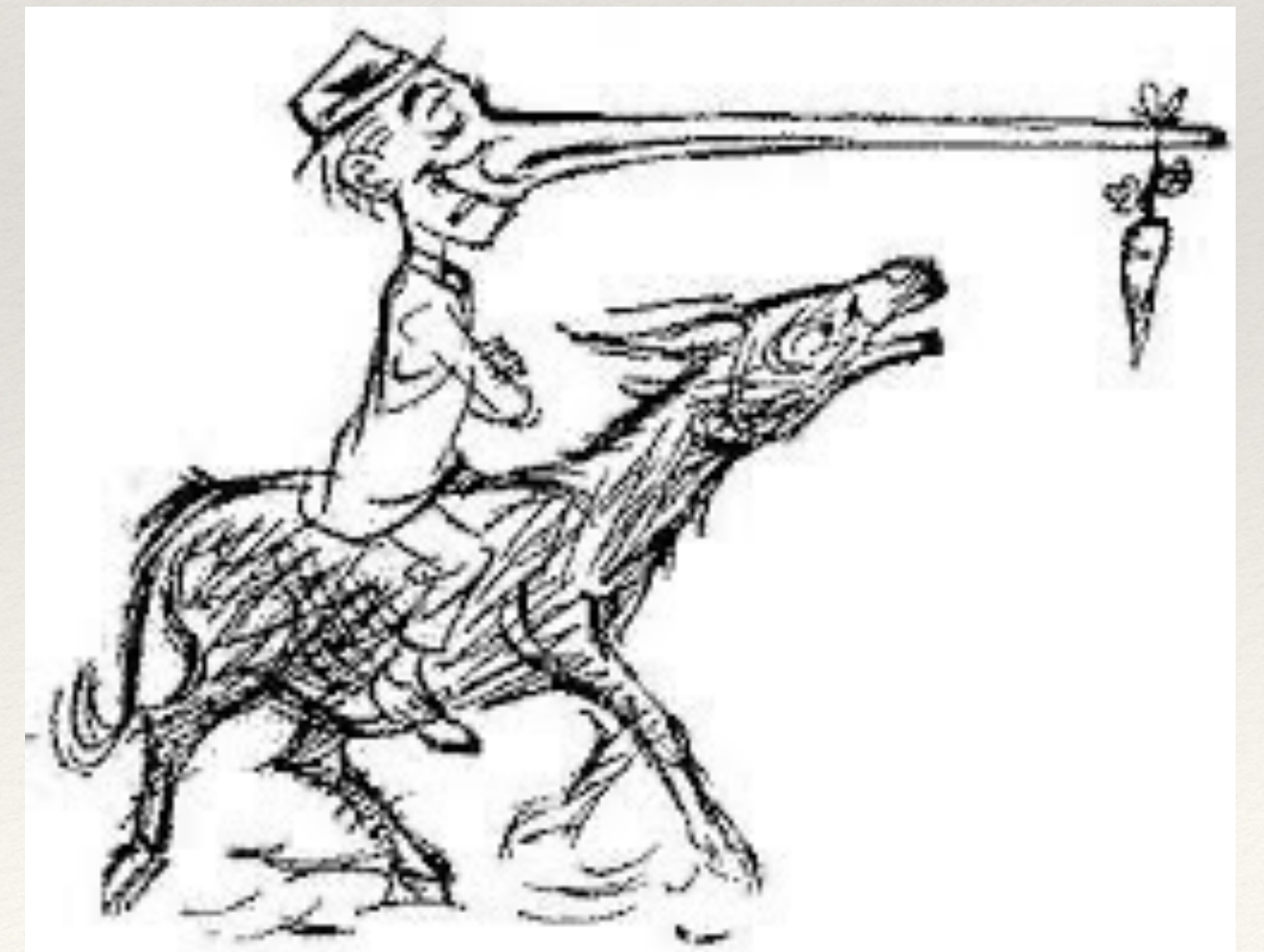
Quantum Mechanics

Lecture 3

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Recap

- Wave particle duality: waves showing particle nature and particle showing wave nature.
- Uncertainty principle, or the lack of precise simultaneous measurement of position and momentum in the quantum world marks a significant departure from the deterministic approach of classical mechanics.
- We are in search of a formalism which can consistently address these issues built on a probabilistic approach.



Wave packet: Particle as Wave

A particle is localised in space, but a wave is spread.

Q: How can I achieve a particle like behaviour from a wave ?

Several waves superposed may interfere constructively in a small region giving the resultant wave a large amplitude and interfere destructively far away that the resultant wave has a small amplitude away, providing a local particle like response.

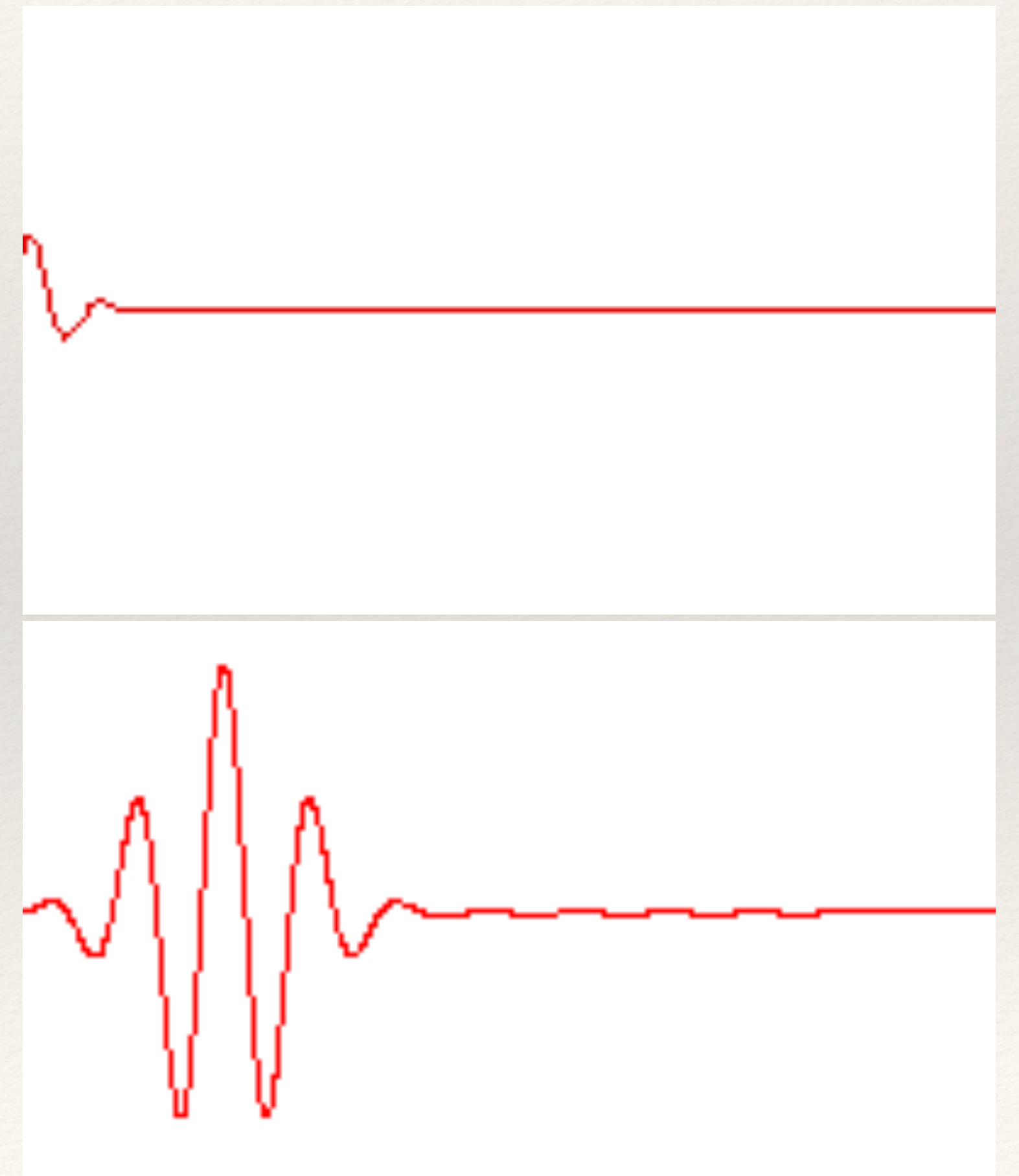
Consider two
sinusoidal waves
Superposing

$$k_2 = k_1 + \Delta k$$

$$\begin{aligned} y &= A\cos(k_1x) + A\cos(k_2x) \\ &= 2A\cos\left(\frac{\Delta k}{2}x\right)\cos\left(\frac{k_1 + k_2}{2}x\right) \end{aligned}$$

● $\cos\left(\frac{\Delta k}{2}x\right) \rightarrow$ specifies an envelope

● $\cos\left(\frac{k_1 + k_2}{2}x\right) \rightarrow$ variation of amplitude of the resultant within the envelope



Wave packet: Example

Consider, several waves superposed:

With discrete wave number (k):

$$y(x) = \sum_{k_i} A(k_i) e^{ik_i x}$$

With continuous wave number (k):

$$y(x) = \int_{-\infty}^{\infty} A(k) e^{ikx} dk$$

Note: The amplitude is a function of 'k'

Consider: $A(k) = e^{-\alpha(k-k_0)^2/2} \rightarrow \Delta k = 2/\sqrt{\alpha}$

Then, $y(x) = \int_{-\infty}^{\infty} e^{-\alpha(k-k_0)^2/2} e^{ikx} dk$

Substitute: $q = k - k_0$ A little algebra:

$$y(x) = e^{ik_0 x} \int_{-\infty}^{+\infty} e^{-\alpha(\frac{q}{\sqrt{2}} - \frac{ix}{(\sqrt{2}\alpha)})^2} e^{-x^2/(2\alpha)} dq$$

Substitute, $\frac{q}{\sqrt{2}} - \frac{ix}{(\sqrt{2}\alpha)} = q' \rightarrow dq = \sqrt{2} dq'$

$$y(x) = \sqrt{2} e^{ik_0 x} e^{-x^2/(2\alpha)} \int_{-\infty}^{+\infty} e^{-\alpha q'^2} dq'$$

$$= \sqrt{\frac{2\pi}{\alpha}} e^{ik_0 x} e^{-x^2/(2\alpha)} \int_{-\infty}^{+\infty} dk e^{-\alpha k^2} = \sqrt{\frac{\pi}{\alpha}}$$

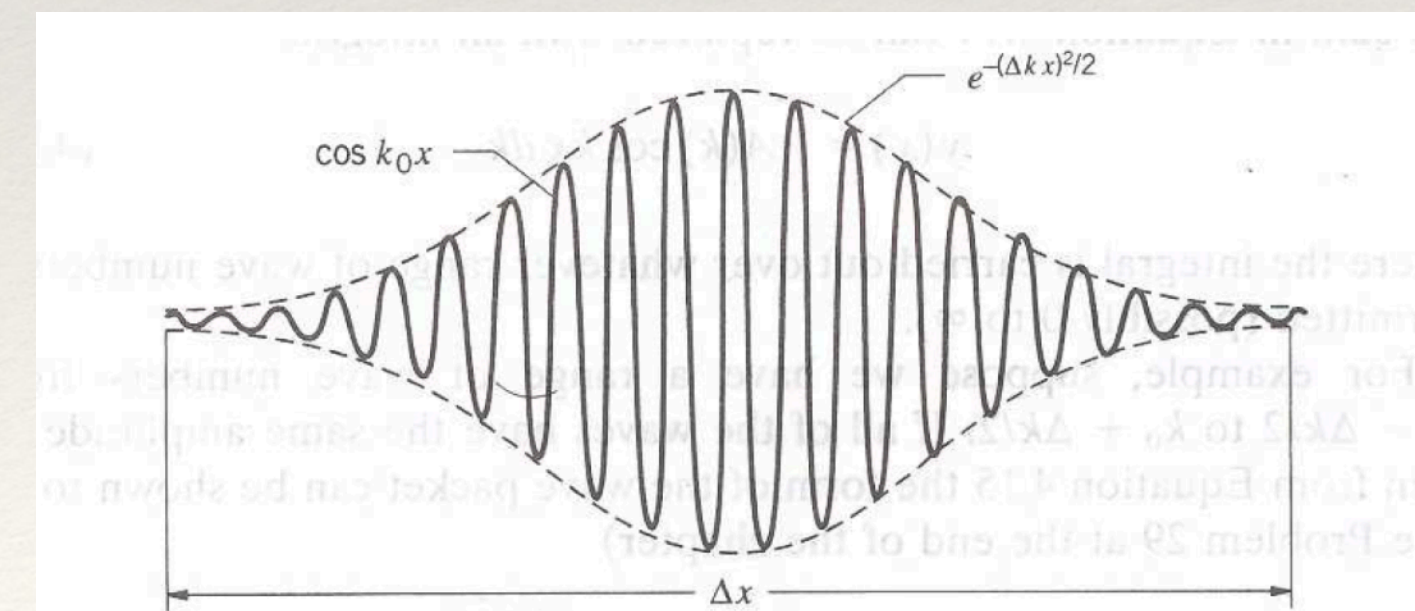
Represents a Wave packet

$$\Delta x = 2\sqrt{\alpha}$$

- $e^{-x^2/(2\alpha)}$ represents the falling behaviour of the wave packet.

$$\Delta x \Delta k = 4$$

- $e^{ik_0 x}$ indicates oscillation within envelop



But the wave/
wave packet is
stationary!

Consider: Travelling Waves

- Consider two waves travelling along x - direction with velocities $v_1 = \omega_1/k_1$ and $v_2 = \omega_2/k_2$. They are :

$$\begin{aligned} y_1(x, t) &= A \cos(k_1 x - \omega_1 t) \\ y_2(x, t) &= A \cos(k_2 x - \omega_2 t) \end{aligned}$$

The resultant wave is: $y(x, t) = y_1 + y_2$

$$y(x, t) = 2A \cos\left(\frac{\Delta k}{2}x - \frac{\Delta \omega}{2}t\right) \cos\left(\frac{k_1 + k_2}{2}x - \frac{\omega_1 + \omega_2}{2}t\right)$$

where, $\Delta \omega = \omega_2 - \omega_1$.

$$\Delta k = k_2 - k_1$$

Envelop (shape of the wave packet)

Motion of the waves within the envelop

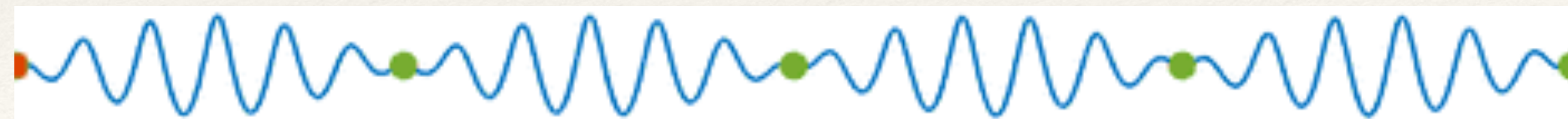
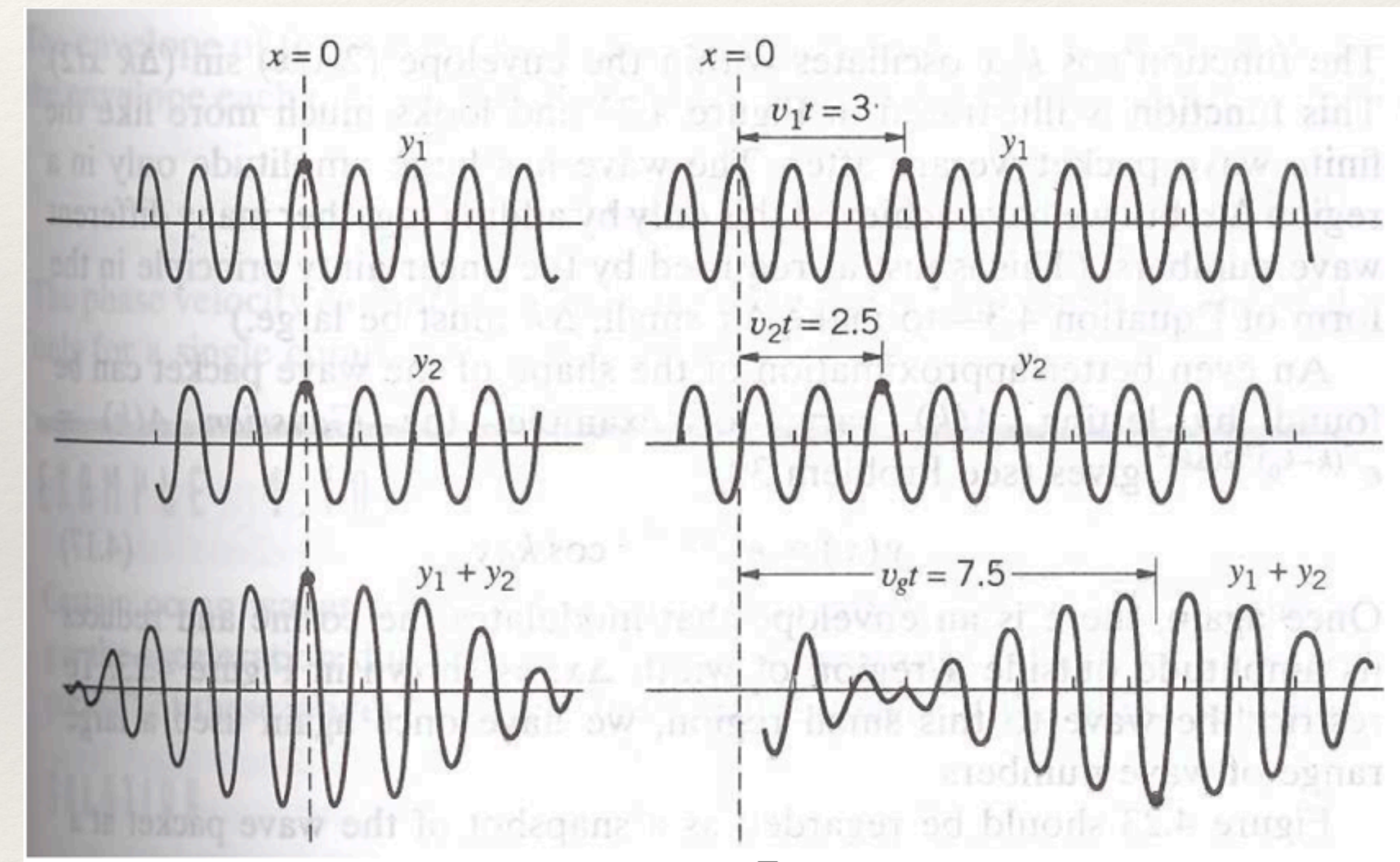
Velocity of the waves within envelop: Phase velocity

$$v = \frac{\omega_1 + \omega_2}{k_1 + k_2}$$

Velocity of the wave packet: Group velocity

$$v_g = \frac{\Delta \omega}{\Delta k}$$

$$y_1 = 1, \quad y_2 = 10/9, \quad v_1 = 3, \quad v_2 = 2.5$$



Travelling particle wave

For matter wave:

$$\omega = \omega(k)$$

Travelling
particle
wave:

$$y(x, t) = \int_{-\infty}^{+\infty} A(k) e^{i(kx - \omega(k)t)} dk$$

Consider a wave packet strongly localized in k-space about a value k_0 .
Therefore, we can expand $\omega(k)$ about k_0 :

$$\omega(k) \approx \underbrace{\omega(k_0)}_{\text{Constant}} + \underbrace{(k - k_0) \left(\frac{d\omega}{dk} \right)_{k_0}}_{\text{Propagation of wave packet}} + \underbrace{\frac{1}{2} (k - k_0)^2 \left(\frac{d^2\omega}{dk^2} \right)_{k_0}}_{\text{Damping}}$$

Consider,

$$v_g = \left(\frac{d\omega}{dk} \right)_{k_0}$$

$$\beta = \left(\frac{d^2\omega}{dk^2} \right)_{k_0}$$

Assume,

$$A(k) = e^{-\alpha(k-k_0)^2}$$

$$k - k_0 = k'$$

$$y(x, t) = e^{i(k_0 x - \omega(k_0)t)} \int_{-\infty}^{+\infty} dk' e^{-\alpha k'^2} e^{ik'(x - v_g t)} e^{-ik'^2 \beta t/2}$$

Again, using,

$$\int_{-\infty}^{+\infty} dk e^{-\alpha k^2} = \sqrt{\frac{\pi}{\alpha}}$$

$$y(x, t) = e^{i[k_0 x - \omega(k_0)t]} \left(\frac{\pi}{\alpha + i\beta t} \right)^{1/2} e^{-[(x - v_g t)^2 / 4(\alpha + i\beta t)]}$$

$$|y(x, t)|^2 = \left(\frac{\pi^2}{\alpha^2 + \beta^2 t^2} \right)^{1/2} e^{-[\alpha(x - v_g t)^2 / 2(\alpha^2 + \beta^2 t^2)]}$$

Represents a wave packet

Therefore, the wave packet's peak is travelling with velocity v_g and it's width varies. Width is proportional to

$$(\alpha^2 + \beta^2 t^2)^{1/2}$$

Phase velocity and group velocity

There are two velocities that are identified in a Wave packet

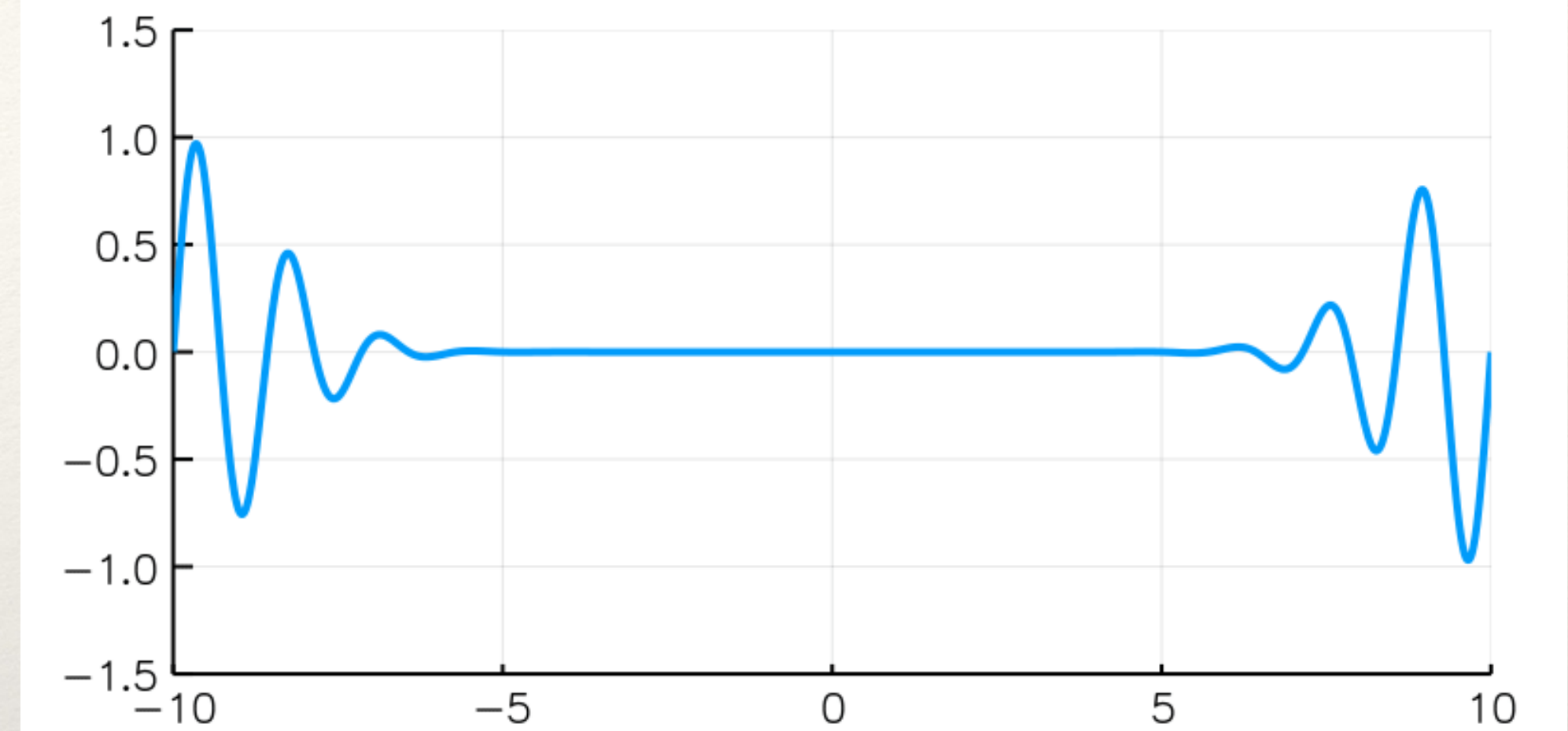
$$y(x, t) = \int_{-\infty}^{+\infty} A(k) e^{i(kx - \omega t)} dk$$

Velocity of the waves within the envelop, Phase velocity

$$v_p = \frac{\omega}{k},$$

Velocity of the Envelop, Group Velocity

$$v_g = \frac{d\omega}{dk}$$



$$v_g = \frac{d\omega}{dk} = \left(\frac{d\omega}{dE} \right) \left(\frac{dE}{dp} \right) \left(\frac{dp}{dk} \right) = \frac{dE}{dp} = v$$

Classical particle

$$E = \frac{p^2}{2m}, \Rightarrow \frac{dE}{dp} = \frac{p}{m} = v$$

Assuming,

$$E = \hbar\omega$$

$$k = \frac{2\pi}{\lambda} = \frac{p}{\hbar}$$

$$\Rightarrow \frac{dE}{d\omega} = \frac{dp}{dk}$$

Group velocity represents the velocity of particle

Wave function

- For matter waves, wave packet describes the motion of the matter completely in terms of a single function $\psi(x, t)$

All informations about the particle is contained in a single function: Wave function $\psi(x, t)$

Wave packet:

$$y(x, t) = \int_{-\infty}^{+\infty} A(k) e^{i(kx - \omega t)} dk$$

In terms of p and E

$$E = \hbar\omega$$

$$k = \frac{2\pi}{\lambda} = \frac{p}{\hbar}$$

Wave function:

$$\psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \int dp \phi(p) e^{i(px - Et)/\hbar}$$

Let's not worry about it now!

❖ Question: What is the dynamics ?

Recall,

$$\Delta x \Delta k = 4$$



$$\Delta x \Delta p = 4\hbar$$

Uncertainty principle !

Partial derivative of wave function with respect to time:

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = \frac{1}{\sqrt{2\pi\hbar}} \int dp \phi(p) E e^{i(px - Et)/\hbar}$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int dp \phi(p) \left[\frac{p^2}{2m} \right] e^{i(px - Et)/\hbar}$$

$$= -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2}$$

The wave function is a general solution of the partial differential equation:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} = i\hbar \frac{\partial \psi(x, t)}{\partial t}$$

For free particle

The equation is a first order in time derivative.

If $\psi(x, 0)$ is known, then $\psi(x, t)$ can be found out

Schrodinger Equation

Generalise: For a particle in a Potential $V(x)$

The wave function satisfies a generic equation:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x)\psi(x, t) = i\hbar \frac{\partial \psi(x, t)}{\partial t}$$

Classical Analogue:

$$\frac{p^2}{2m} + V = E$$

In three dimension:

$$\frac{\partial^2}{\partial x^2} \rightarrow \nabla^2$$

We can't derive it !!

The diagram shows the 3D Schrodinger equation:
$$-\frac{\hbar^2}{2m} \nabla^2 \Psi + V\Psi = i\hbar \frac{\partial \Psi}{\partial t}$$
 with the following annotations:

- $\hbar$: Planck's constant
- ∇^2 : Called the 'del-squared operator', this quantity describes how the wavefunction, Ψ , changes from one place to another
- i : A mathematical quantity called an 'imaginary number'. It is equal to the square root of minus one
- m : The mass of the particle being described
- V : Describes the forces acting on the particle
- $\frac{\partial \Psi}{\partial t}$: Describes how Ψ changes its shape with time



Erwin Schrödinger
Nobel prize: 1933

Interpretation

What does the wave function imply ?

- Max Born (1926): Probability that the particle is found between x and $x + dx$ at time t is given by

$$P(x, t)dx = |\psi(x, t)|^2 dx$$

Statistical
interpretation of
wave function

Probability density of finding the particle: $|\psi(x, t)|^2$

However, the total probability of finding the particle in the whole space must be conserved.

$$\int_{-\infty}^{\infty} P(x, t) dx = 1$$

➔ $\int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = 1$

Chooses a class of functions that are **square integrable** !

Normalisation of wave function

Max Born
Nobel prize: 1954

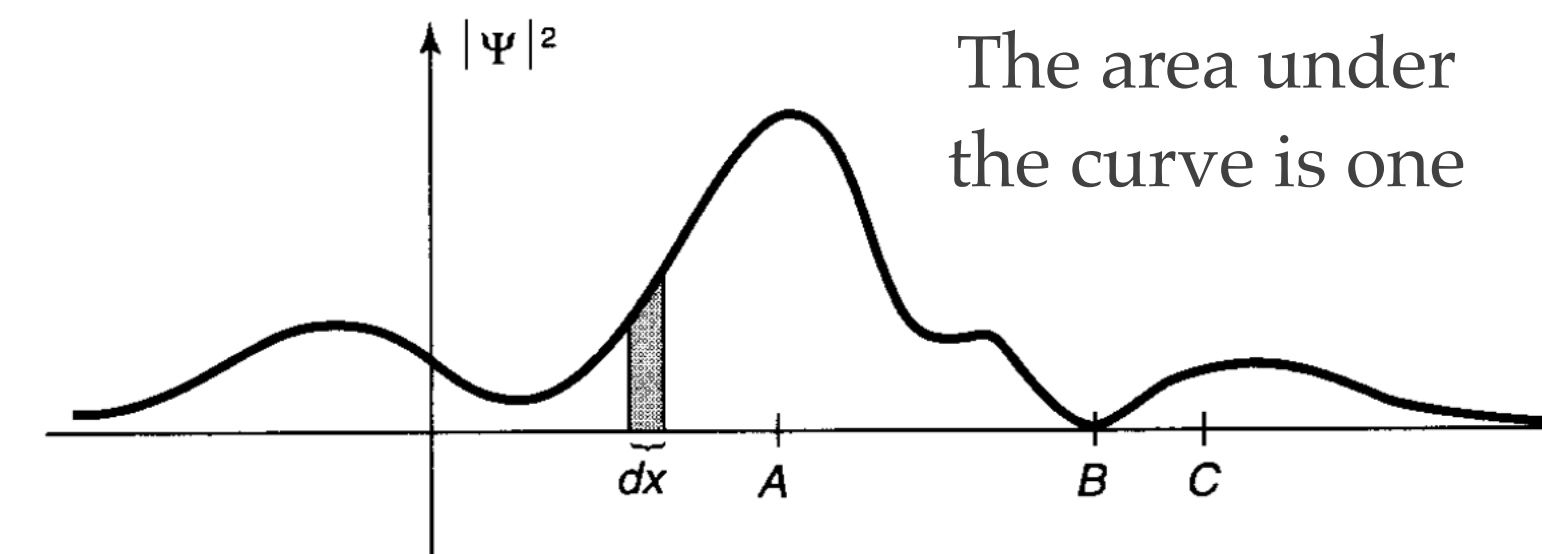


Q: But, suppose I normalise the wave function at $t=0$. How does it guarantee that it will remain normalisable at any time ?

$$\frac{d}{dt} \int_{-\infty}^{+\infty} |\psi(x, t)|^2 dx = 0$$

The whole integral remains time independent !

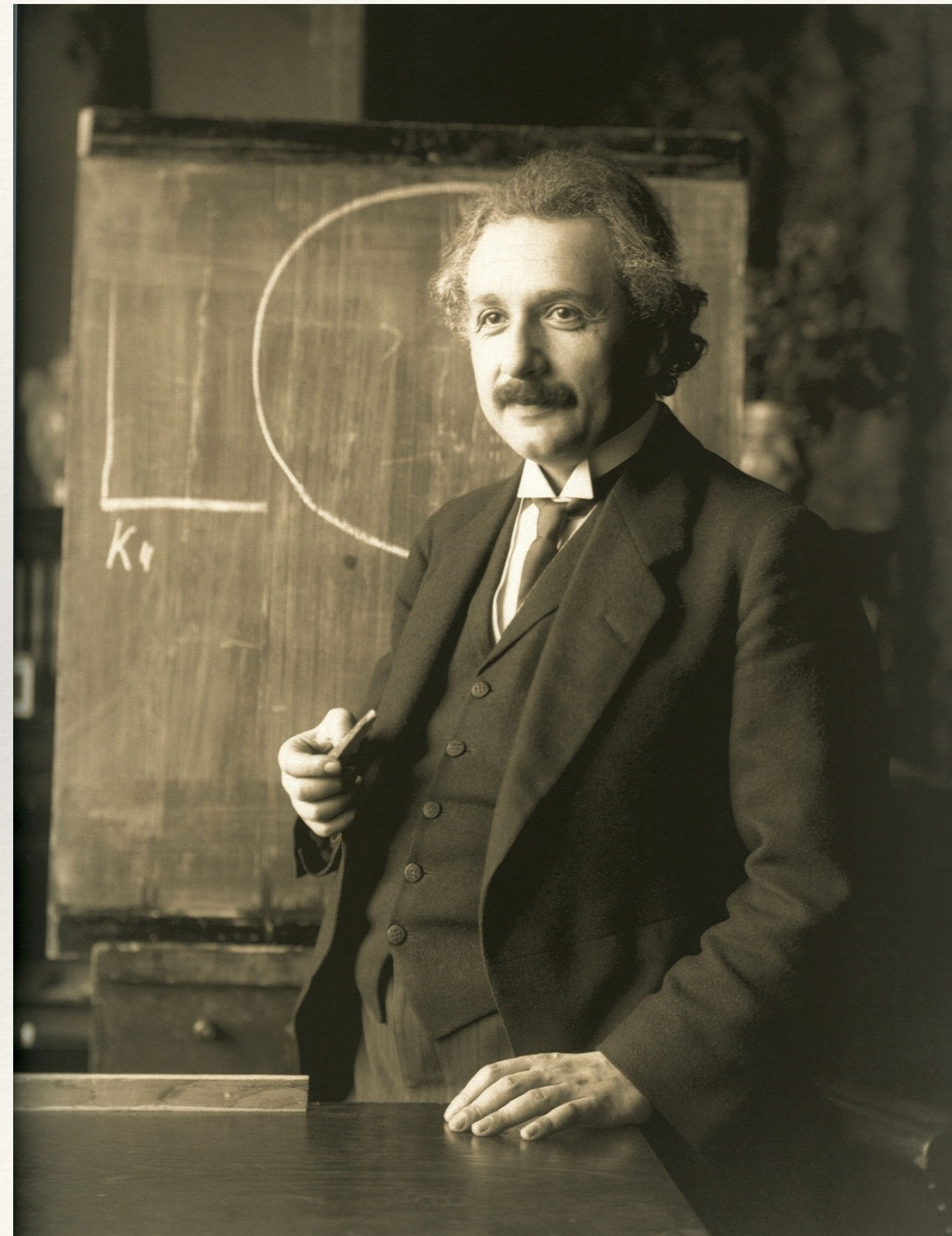
- $\psi(x, t)$ must be continuous and single valued.



Einstein on Modern Quantum Theory

Einstein didn't like the statistical interpretation of quantum mechanics

- The more success the quantum theory has, the sillier it looks.
- If quantum theory is correct, it signifies the end of physics as a science.
- God does not play dice.



Summary

The concept and construct of wave packet does address:

Wave particle duality

Uncertainty principle

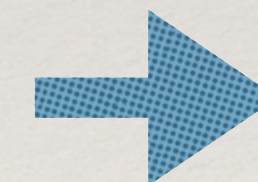
Dynamics of a particle

Schrödinger Equation
for wave functions:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x)\psi(x, t) = i\hbar \frac{\partial \psi(x, t)}{\partial t}$$

$$|\psi(x, t)|^2$$

Provides probability density



$$\int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = 1$$

Normalisation: A necessary condition
for the wave functions to satisfy